

作 8

$$\begin{aligned}
 & \frac{1}{4\pi^2} \iint_{R^2} F(u,v) e^{j(ux+vy)} du dv \\
 &= \frac{1}{4\pi^2} \iint_{R^2} \left(\iint_{R^2} f(m,n) e^{-j(um+vn)} dm dn \right) e^{j(ux+vy)} du dv \\
 &= \frac{1}{4\pi^2} \iint_{R^2} f(m,n) dm dn \iint_{R^2} e^{ju(x-m)} \cdot e^{jv(y-n)} du dv \\
 &= \frac{1}{4\pi^2} \iint_{R^2} f(m,n) dm dn \cdot \int_{-\infty}^{+\infty} e^{ju(x-m)} du \cdot \int_{-\infty}^{+\infty} e^{jv(y-n)} dv
 \end{aligned}$$

已知

$$1 \xrightarrow{F} 2\pi \delta(\omega)$$

$$\text{即} \int_{-\infty}^{+\infty} e^{-j\omega t} dt = 2\pi \delta(\omega)$$

$$\text{用 } \omega' = -\omega$$

$$\int_{-\infty}^{+\infty} e^{j\omega' t} dt = 2\pi \delta(-\omega') = 2\pi \delta(\omega')$$

$$\text{即} \int_{-\infty}^{+\infty} e^{j\omega t} dt = 2\pi \delta(\omega)$$

$$\text{所以} \int_{-\infty}^{+\infty} e^{ju(x-m)} du = \delta(x-m) \cdot 2\pi$$

$$\int_{-\infty}^{+\infty} e^{jv(y-n)} dv = \delta(y-n) \cdot 2\pi$$

$$\text{原式} = \iint_{R^2} f(m,n) dm dn \cdot \delta(x-m) \cdot \delta(y-n)$$

$$= f(x,y) \iint_{R^2} \delta(m-x) \cdot \delta(n-y) dm dn$$

$$= f(x,y)$$

$$\therefore f(x,y) = \frac{1}{4\pi^2} \int_{-\infty}^{+\infty} \int_{-\infty}^{+\infty} F(u,v) e^{j(ux+vy)} du dv$$